

Scaling medical imaging report generation with multimodal reinforcement learning

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Abstract

Frontier models have demonstrated remarkable capabilities in understanding and reasoning with natural-language text, but they still exhibit major competency gaps in multimodal understanding and reasoning especially in high-value verticals such as biomedicine. Medical imaging report generation is a prominent example. Supervised fine-tuning can substantially improve performance, but they are prone to overfitting to superficial boilerplate patterns. In this paper, we introduce Universal Report Generation (UNIRG) as a general framework for medical imaging report generation. By leveraging reinforcement learning as a unifying mechanism to directly optimize for evaluation metrics designed for end applications, UNIRG can significantly improve upon supervised fine-tuning and attain durable generalization across diverse institutions and clinical practices. We trained UNIRG-CXR on publicly available chest X-ray (CXR) data and conducted a thorough evaluation in CXR report generation with rigorous evaluation scenarios. On the authoritative ReXrank benchmark, UNIRG-CXR sets new overall SOTA, outperforming prior state of the art by a wide margin.

Main

Medical imaging report generation has been an important application area for medical foundation models, aiming to automatically produce coherent and clinically meaningful diagnostic reports from medical images such as chest radiographs. Beyond its potential to reduce reporting burden, improve workflow efficiency, and enhance diagnostic consistency, report generation represents a key benchmark for evaluating broader multimodal reasoning capabilities in healthcare AI. Despite recent progress driven by large-scale vision–language models and domain-specific training corpora [17, 26, 32], enabling models to produce faithful, clinically grounded reports that generalize across real-world imaging environments remains a substantial challenge.

A central obstacle lies in cross-institution generalization. For example, radiology reporting practices vary widely across datasets and healthcare systems, influenced by differences in institutional guidelines, departmental conventions, radiologist writing styles, and patient populations [5]. Consequently, models trained through supervised fine-tuning (SFT) tend to inherit the lexical biases, phrasing patterns of their training datasets. While such models may achieve high scores on in-distribution benchmarks, they often show substantial degradation when evaluated on unseen institutions or external datasets. This brittleness is particularly concerning for safety-critical applications, where reliable performance across demographic groups, institutions, and imaging conditions is essential.

A second major limitation of prior work is that SFT objectives primarily optimize next-word prediction, which encourages surface-level lexical similarity to reference reports rather than alignment with clinically important factual attributes. As a result, conventional report generation systems often overfit to n-gram–based metrics (e.g., BLEU, ROUGE) that are only weakly correlated with radiological correctness [22, 26]. This misalignment highlights the need for training paradigms that optimize models directly for clinical usefulness and factual accuracy rather than superficial linguistic similarity.

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To address these challenges, we introduce UNiRG, a unified and universal report generation framework built upon a novel reinforcement learning framework designed to enhance generalization, factual alignment, and robustness. Our RL strategy directly optimizes clinically grounded reward signals that capture disease-specific correctness, structured factual relationships, and multi-metric performance. Using this framework, we train UNiRG-CXR on chest X-rays and show that it moves beyond dataset-specific reporting conventions, instead learning generalizable representations that yield consistently strong performance across diverse clinical settings.

We conduct extensive and rigorous evaluations to benchmark UNiRG-CXR against existing radiology report generation models across a broad spectrum of settings. Our evaluation suite includes standard report-level generation metrics, condition/disease-level classification, cross-dataset/institution/demographics generalization, and longitudinal evaluation reflecting real-world clinical practices. Across all axes, UNiRG-CXR achieves overall state-of-the-art (SOTA) performance, consistently surpassing prior baselines and demonstrating universal, reliable capabilities rarely observed in prior RRG systems.

In summary, our contributions are threefold:

A new reinforcement learning framework (UniRG) for radiology report generation, yielding UNiRG-CXR, a single model that attains overall SOTA performance across multiple datasets and evaluation metrics.

A comprehensive, clinically grounded evaluation of UniRG-CXR, spanning report-level metrics, disease-level correctness, cross-dataset and cross-institution generalization, robustness analyses, and longitudinal assessment.

Demonstration of UniRG-CXR’s universal capabilities, showing that UNiRG-CXR overcomes the longstanding specialization and metric-overfitting issues of SFT-based report generation models by producing clinically aligned, generalizable reports across diverse real-world conditions.

Together, these advances establish UniRG as a robust and generalizable foundation for radiology report generation and highlight the promise of reinforcement learning as a key component of next-generation clinical vision–language models.

Results

Overview of UniRG and UniRG-CXR

We propose UNiRG, a next-generation approach for medical imaging report using reinforcement learning. With this framework, we train UNiRG-CXR, a state-of-the-art model for radiology report generation that produces clinically faithful reports across diverse institutions and datasets with substantial lead over existing baselines. Built upon the open-source Qwen3-VL-8B-Instruct foundation, UNiRG-CXR combined supervised fine-tuning with reinforcement learning to directly optimize for clinically relevant objective aggregating multiple metrics spanning rule-based (BLEU), model-based (BERTScore, SemScore, RadGraph F1), and LLM-based (CheXprompt) metrics. Trained on large-scale datasets including MIMIC-CXR, CheXpert Plus, ReXGradient and IU (covering over 560k studies from 80+ institutions), UNiRG-CXR can condition on the current image, contextual text, and prior study information where available and output both findings and impression as a full radiology report from the x-ray interpretation. To evaluate the quality of UNiRG-CXR, we assess UNiRG-CXR on held-out test sets (MIMIC-CXR [11], CheXpert Plus [4], ReXGradient [30], IU [6]) and unseen proprietary data. Report quality is measured using ReXrank metrics [29] and CheXprompt, an LLM-based clinical-error metric [27], while diagnostic ability is evaluated via F1-based disease classification from generated reports. UNiRG-CXR demonstrates unprecedented generalization and robustness, outperforming prior state-of-the-art systems (e.g., MedVersa, MedGemma, MAIRA-2) by substantial margins across public and private benchmarks across our evaluation settings.

As shown in (d) and (e) from Figure 2, we present the detailed ReXrank leaderboard results¹ which compares UNiRG-CXR against previous SOTA models across four datasets—ReXGradient [30], MIMIC-CXR [11], IU X-Ray [6], and CheXpert Plus [4]. Each dataset is evaluated under two generation settings: findings-only and the more challenging findings + impression setup. Performance is measured by 1/RadCliQ-v1 [24] (higher is better), a composite metric used as the default metric in RexRank leaderboard and was

¹Results are collected from the ReXrank leaderboard [29]

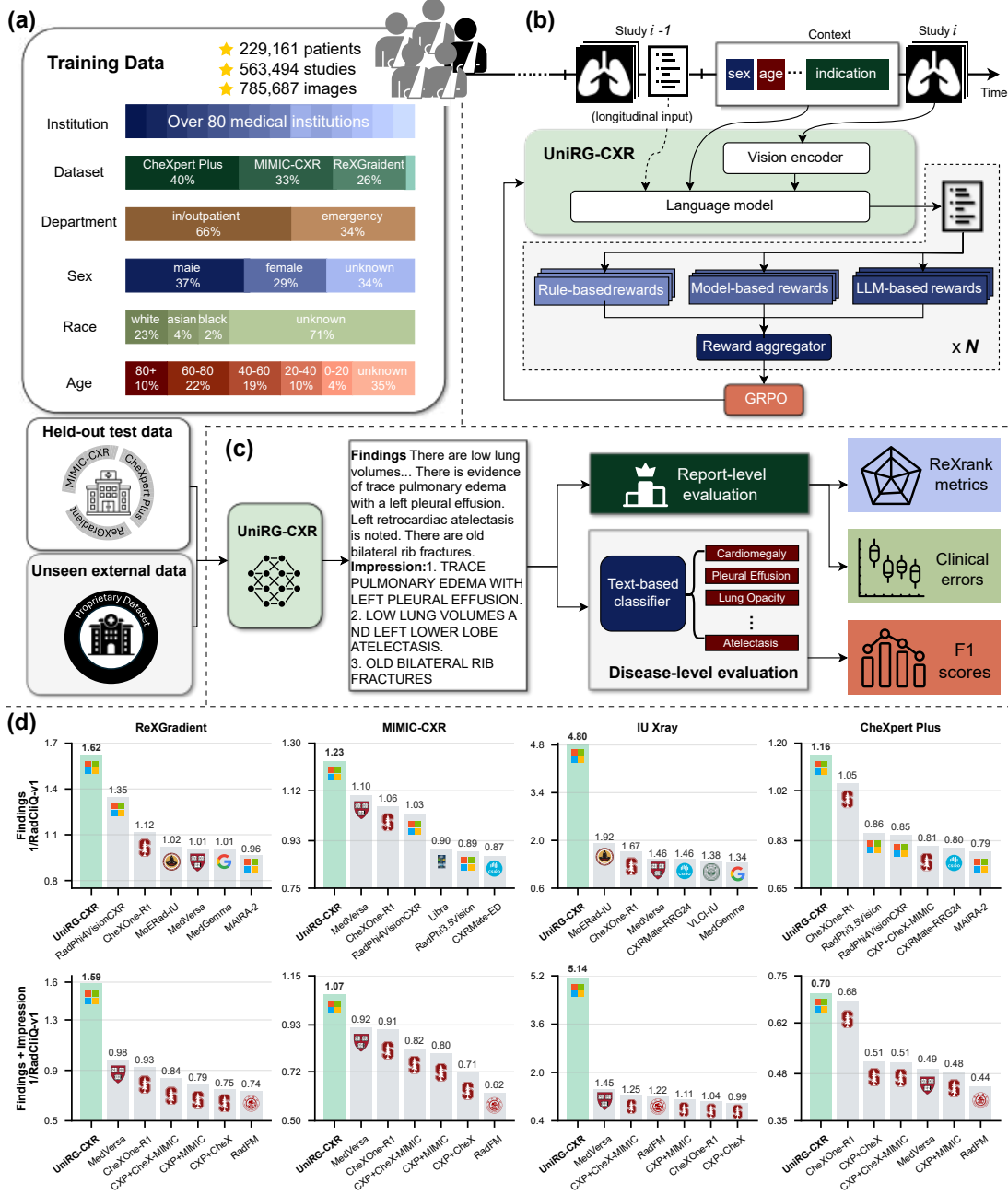


Figure 1: Overview of UniRG-CXR. (a) Training Data: UniRG-CXR is trained on the training splits of MIMIC-CXR [11], CheXpert Plus [4], ReXGradient-160k [30] and IU [6] covering diverse institutions and patient demographics. (b) Training and Rewards: Taking input from the current image, clinical context (e.g., indication), and optionally prior studies, UniRG-CXR uses GRPO reinforcement learning to optimize composite rewards that combine rule-based, model-based, and LLM-based metrics. (c) Evaluation: We assess UniRG-CXR on held-out test sets from MIMIC-CXR, CheXpert Plus, ReXGradient and additionally assess zero-shot generalization on an external proprietary dataset and IU-Xray (the zero-shot setting excludes IU-Xray training data). Report quality is measured using ReXrank metrics [29] and an LLM-based clinical-error metric [27], while diagnostic ability is evaluated via F1-based disease classification from generated reports. (d) ReXrank Results: UniRG-CXR achieves SOTA performance across four datasets and two generation settings (findings only and findings + impression), showing substantial gains over prior state-of-the-art.

found to correlate more strongly with human judgment than individual metrics. As shown in Figure 1, UNiRG-CXR achieves consistent and substantial gains over prior SOTA models in every steup. Notably, on the ReXGradient and IU test set, UNiRG-CXR exceeds the previous best model by over 50%, underscoring its strong generalization capability for radiology report generation.

UniRG-CXR achieves SOTA with universal improvements across metrics

While RadCliQ-v1 serves as our primary evaluation metric, Figure 2 further illustrates that UNiRG-CXR delivers broad improvements across diverse metrics including both those emphasizing lexical similarity (e.g., BLEU, BERTScore) and those emphasizing factual correctness (e.g., SembScore). Notably, UNiRG-CXR also improves metrics it was not explicitly optimized for, such as RaTEScore, highlighting that its performance gains are universal and multi-faceted rather than overfitted to specific metrics (Figure 2 (a)). The key to this universal improvements lies in our combined reward RL which jointly optimizes multiple individual metrics. As shown in Figure 2 (b) and (c), we compare UNiRG-CXR with ablation studies that only optimize an individual metric. We show that our proposed combined reward RL achieves balanced gains across individual metrics and yields overall the best RadCliQ-v1 score.

Beyond the ReXrank-provided metrics, we additionally evaluate UNiRG-CXR using CheXprompt [27], an LLM-based metric that quantifies the number of clinical errors relative to the reference report (following Zambrano Chaves et al. [27], we use GPT-4 as the backbone evaluator). As shown in Figure 2 (f), we compare the proportion of generated reports with varying numbers of clinical errors (≤ 1 , 2, 3, and ≥ 4) across UNiRG-CXR, MedVersa, and MedGemma. UNiRG-CXR produces substantially more error-free or low-error reports ($21.3\% \leq 1$ error) compared with prior state-of-the-art systems (Medversa 16.1% and MedGemma 3.1%), while markedly reducing the fraction of high-error reports (≥ 4 errors: 14.8%) relative to MedVersa (32.3%) and MedGemma (43.5%). These results indicate that UNiRG-CXR achieves more clinically faithful and accurate report generation. Qualitatively, we show an example where UNiRG-CXR can generate error-free report that covers all the important findings while output from MedGemma and MedVersa contain errors or partial errors. The key to the improvement on clinical error reduction is the incorporation of error awareness metric (i.e. CheXprompt) in our combined RL recipe in UNiRG-CXR. In Figure 2 (d), the RL(no error awareness) ablation that does not contain CheXprompt optimization shows stagnant learning in terms of reducing clinical errors whereas our full RL recipe from UNiRG-CXR that incorporates error awareness leads to a steady downward trajectory of reduction in report errors throughout training, indicating that explicit optimization for clinical correctness effectively improves report fidelity.

UniRG-CXR enhances longitudinal report generation

In the real-world setting, radiologists routinely reference prior studies (both reports and images) when interpreting the current exam, often noting changes such as whether pneumonia has improved or worsened compared to a previous scan. To better approximate this realistic workflow, we incorporate longitudinal training, enabling the model to condition on both the prior image and prior report, during the RL training. As shown in Figure 3, UNiRG-CXR achieves the best performance in longitudinal report generation compared with prior longitudinal report generation models such as Maira-2 and frontier large language model such as GPT-5. We also notice that longitudinal information is effectively incorporated in UNiRG-CXR as it boosts performance over its non-longitudinal set up. To understand the gain of UNiRG-CXR in more granularity, we further categorize the reports as shown in Figure 3 (b). The test studies from MIMIC are split into five categories according to their encounter time points ranging from first encounter report without prior information to increased complexity in 2nd, 3rd, 4th and 5th+ encounter points where the report is written in reference with multiple encounters the patient has experienced in the history. We observe that the first-encounter reports are generally the most challenging, and as the number of encounters increases, report quality improves consistently. This trend is intuitive: for the first encounter, the model must generate a completely new description based solely on the current image, without any prior context. In contrast, subsequent encounters provide previous reports that capture the patient’s underlying conditions, enabling the model to generate more accurate and contextually faithful reports. Across all encounter points, UNiRG-CXR achieves substantial performance gains over prior models. Moreover, UNiRG-CXR significantly outperforms the “copy prior report” baseline, demonstrating that it effectively leverages prior information rather than

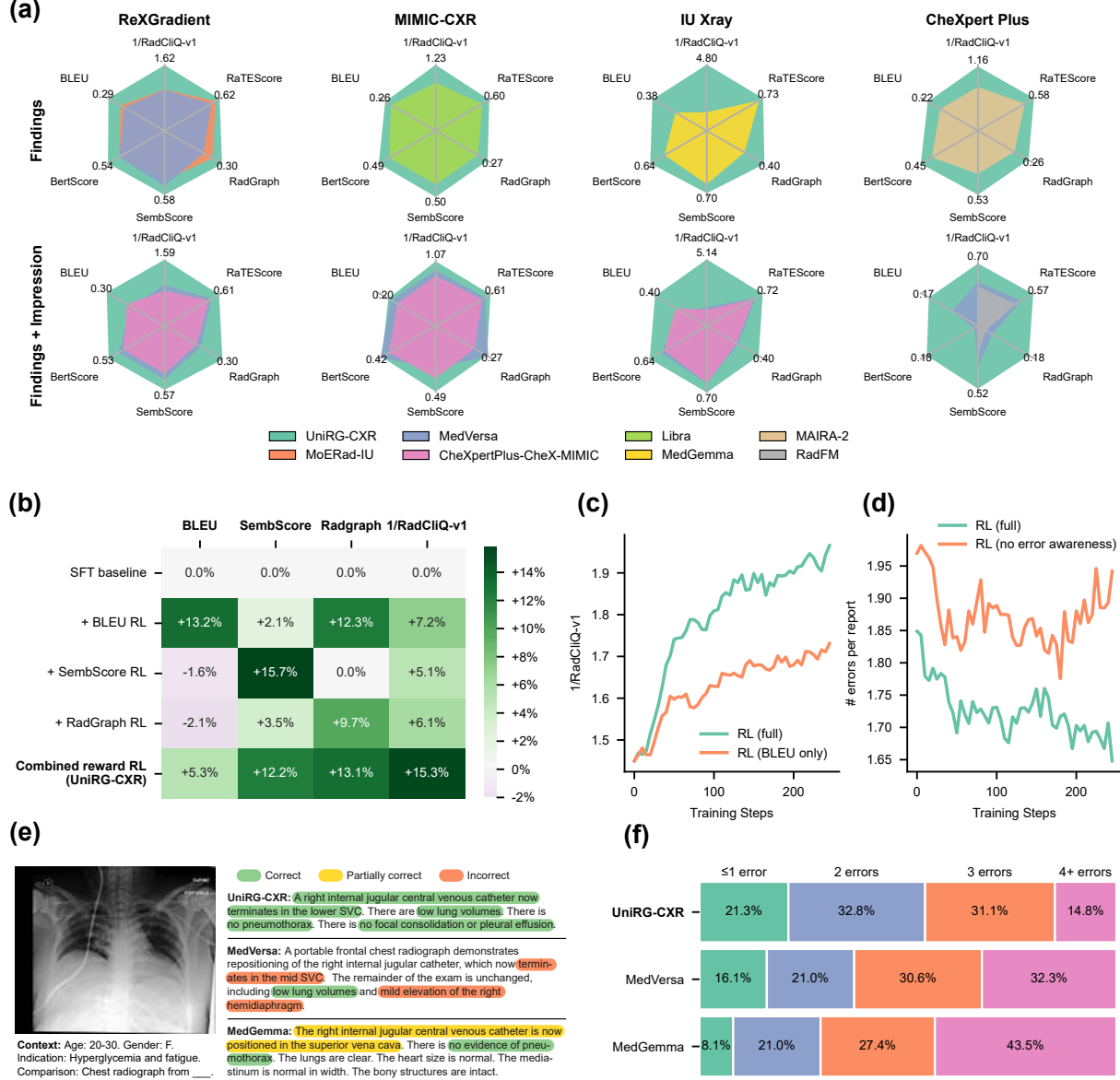


Figure 2: UniRG-CXR achieves state-of-the-art performance, delivering consistent and comprehensive performance gains across metrics. (a) On the ReXrank leaderboard, UniRG-CXR (green) shows robust, universal improvement across all evaluation metrics. (b). Starting from the same SFT checkpoint, RL with our combined reward achieves more balanced gains across metrics and the highest RadCliQ-v1 score compared to RL on single metrics. This ablation study is trained and tested on MIMIC (c). Ablation study on the training dynamics shows RL full (UniRG-CXR) achieves significantly better RadCliQ-v1 score than RL only on BLEU. (d). During training, RL full (UniRG-CXR) shows a steady decrease in clinical errors per report as compared with a fluctuating trajectory without consistent improvement from an ablation run without error awareness (i.e. removing CheXprompt metric optimization). Both (c) and (d) show results on 1024 MIMIC validation set from ablations that are trained on MIMIC. (e). Case studies illustrate that UniRG-CXR can produce error-free reports, unlike MedVersa and MedGemma. (f). UniRG-CXR yields a substantially higher proportion of reports with ≤ 1 error and fewer with ≥ 4 errors than prior models.

relying on it as a shortcut, a behavior observed in some competing models (e.g., MedGemma, GPT-4o, and GPT-5) which only marginally exceed the copy-prior baseline. As longitudinal report generation involves capturing temporal changes in disease state across patient encounters. In Figure 3(c), we categorize each report into five temporal description types (first study (no prior study), new development, no change, progression,

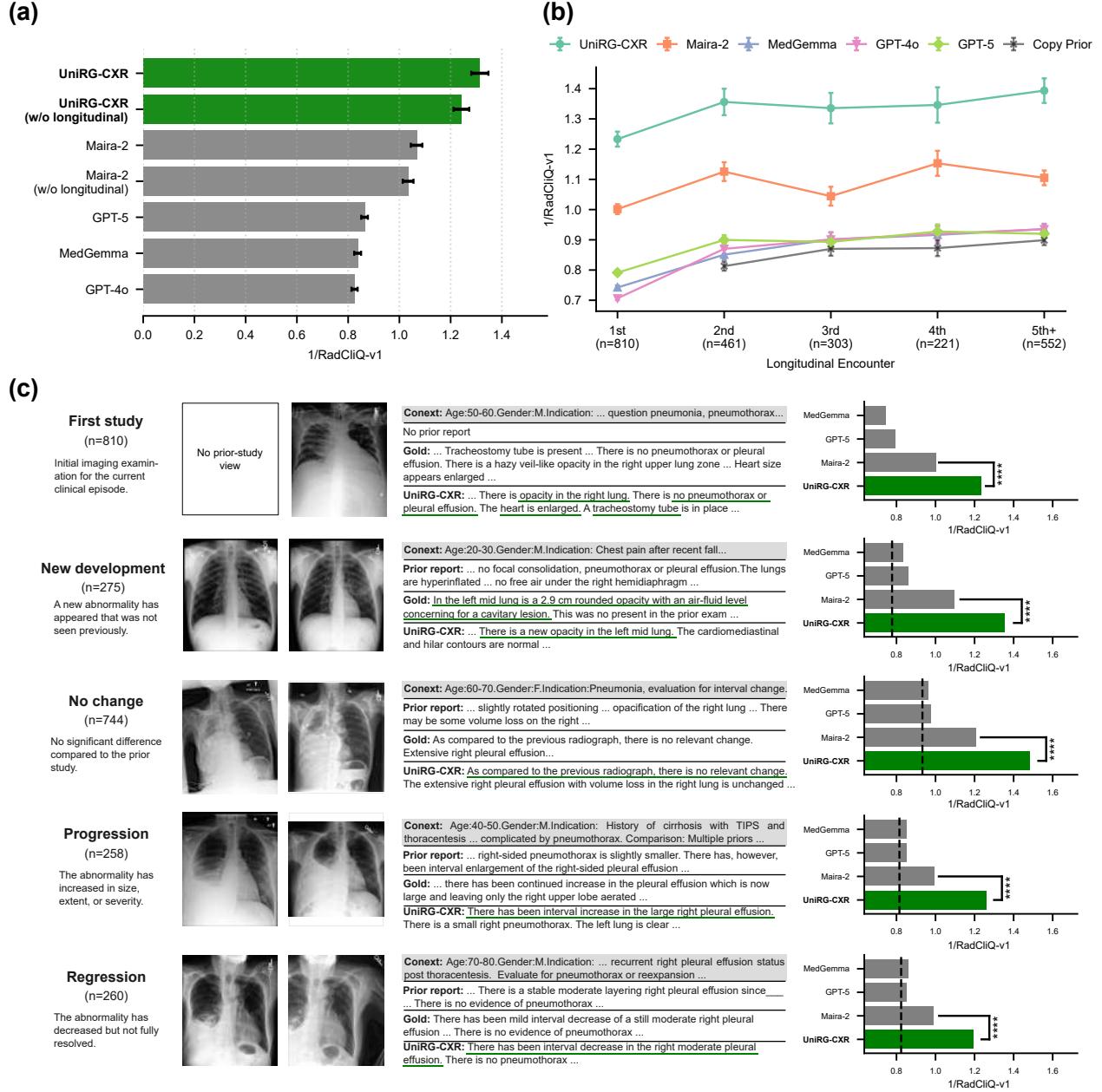


Figure 3: UNIRG-CXR enhances longitudinal report generation. (a). Comparing UNIRG-CXR and its non-longitudinal ablation with prior models on longitudinal report generation, we show UNIRG-CXR exhibits the best performance and the longitudinal information is beneficial to the performance. (b). UNIRG-CXR achieves the best performance across different longitudinal encounter points ranging from the first encounter to the more complex 5th+ encounters, showcasing its improvements are across the board. In comparison, prior models such as GPT-5, GPT-4o and MedGemma are barely surpassing the copy prior report baseline (grey lines). (c). Compared with prior models which barely improve over the copy prior baseline (dashed line), UNIRG-CXR significantly and consistently improves performance across different temporal disease change categories including new development, no change, progression and regression (categorized by GPT-5 on ground truth report). Qualitative examples are shown for each category where UNIRG-CXR correctly predicts the temporal change based on the input. All results in this figure are on MIMIC test set with prior information where available.

and regression) using GPT-5 as an automatic labeler. We evaluate UNIRG-CXR's performance across these categories, comparing it against prior models and the copy prior report baseline. As expected, the no

change category is the easiest, since much of the content can be reused from the prior report. In contrast, categories reflecting disease evolution such as regression or new development are more challenging, as the model must accurately localize and quantify subtle changes. UNiRG-CXR demonstrates consistent and substantial improvements across all categories, markedly surpassing both prior models and the copy-prior baseline. Qualitative examples further illustrate that UNiRG-CXR generates clinically faithful longitudinal descriptions, correctly identifying changes such as new findings or resolving abnormalities. Collectively, these results highlight UNiRG-CXR’s superior capability to model and reason over longitudinal patient trajectories.

Generalization and Robustness of UniRG-CXR

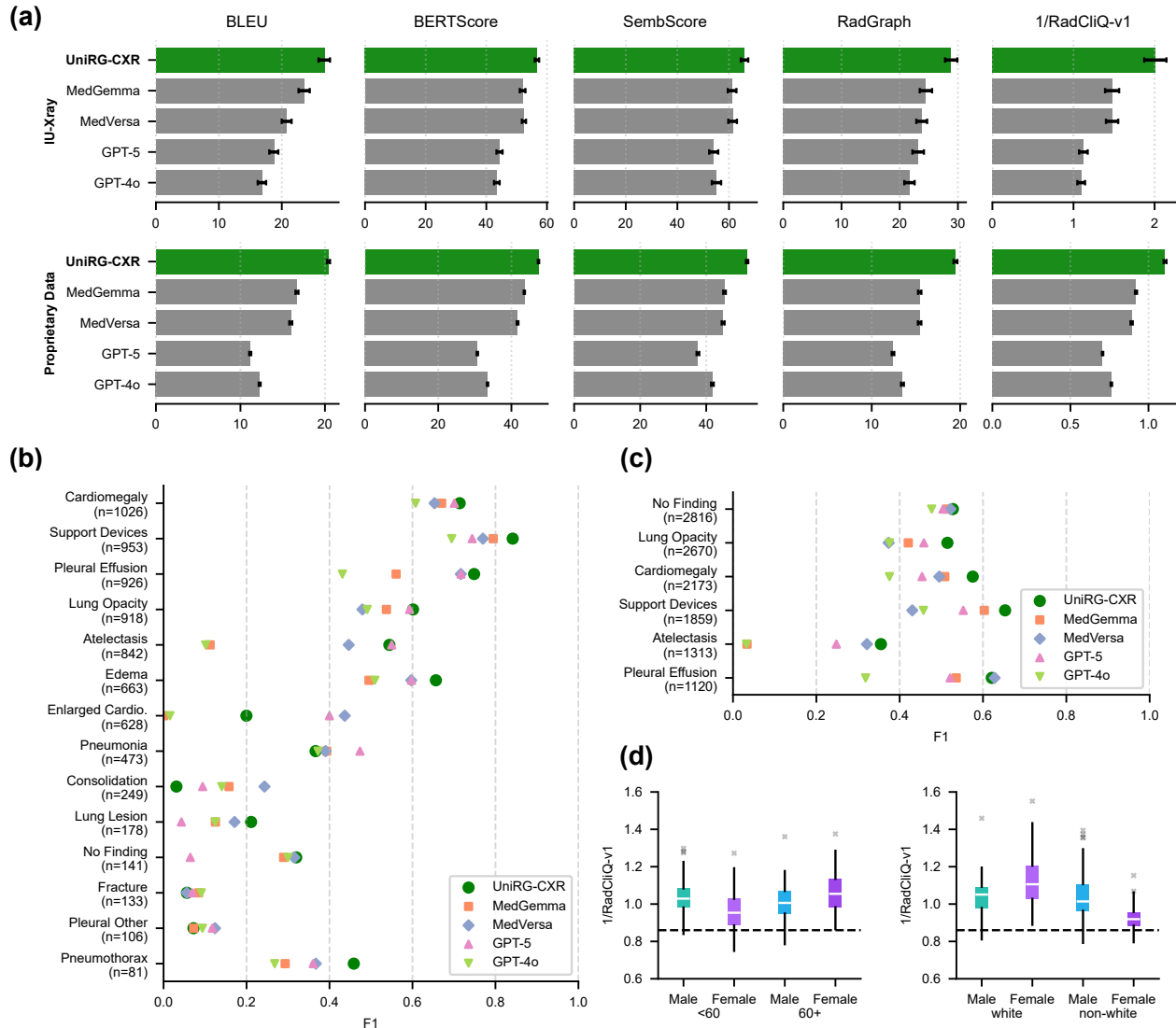


Figure 4: Generalization and robustness of UNiRG-CXR. (a). We held out two datasets sources (IU-Xray and PD (proprietary data) from the training data and evaluate UNiRG-CXR in a zero-shot setting on these datasets. UNiRG-CXR consistently outperforms prior models, maintaining substantial performance gains in this challenging setup. (b) and (c) present condition-level F1 scores on MIMIC-CXR and PD and highlight that UNiRG-CXR remains the overall top-performing model in condition-level diagnostic accuracy. (d). UNiRG-CXR demonstrates stable and robust performance across gender, age, and race subgroups, all of which exceed the performance of the second-best model (the dashed lines).

In this section, we evaluate the robustness and generalization of UNIRG-CXR. First of all, a robust universal report generation model should generalize well across institutions, including those with data distributions unseen during training. To test this, we create an experiment setup where we intentionally leave out certain data sources (IU data and a propriety data source) from the training of UNIRG-CXR. We then test the model’s zero-shot performance on these two datasets, as shown in Figure 4 (a). UNIRG-CXR consistently achieves the best performance across all out-of-distribution datasets, surpassing prior baselines with substantial margins.

Another key generalization of chest radiology report generation models is to accurately identify and classify specific thoracic diseases/conditions, thereby expediting the diagnostic process for clinicians [14]. We evaluate the condition diagnosis capabilities of the output from UNIRG-CXR by applying the CheXbert model [20] to detect diseases from its generated reports. As shown in Figure 4 (b) and (c), UNIRG-CXR is leading the performance across all the prevalent diseases compared with other prior models in both MIMIC and out-of-distribution proprietary dataset.

Finally, we evaluate the robustness of UNIRG-CXR across gender, age, and race. As shown in Figure 4(d), we present performance stratified by these demographic subgroups on the CheXpert-Plus dataset. UNIRG-CXR achieves consistently high scores across all groups, with overlapping distributions and no noticeable performance drop in any demographic category. Moreover, all subgroup performances of UNIRG-CXR surpass those of the second-best model (dashed line), demonstrating that UNIRG-CXR is robust and fair across demographic subgroups while maintaining superiority over prior models.

Discussion

In this work, we introduce UNIRG, a reinforcement learning-based framework for medical imaging report generation. Using this framework, we are able to train UNIRG-CXR, a state-of-the-art radiology report generation system for chest radiographs, which establishes a new performance standard across benchmarks on the ReXrank leaderboard, which spans multiple clinical contexts (inpatient, outpatient, and emergency care) and data from over 70 medical sites. Unlike previous models, which often excelled on isolated benchmarks or on specific metrics, UNIRG-CXR delivers consistent performance improvement across multiple metrics on four widely used datasets (MIMIC-CXR, IU-Xray, CheXpert Plus, and ReXGradient) as well as additional proprietary dataset.

Our evaluation protocol provides a comprehensive assessment of AI-based radiology report generation. We benchmark the system using lexical similarity, embedding-based similarity, and LLM-based clinical error metrics, enabling a holistic comparison against both reference reports and competing models. Beyond these, we conduct longitudinal evaluations that simulate real-world radiologists’ workflows, assess generalization and robustness on unseen distributions, and stratify performance by demographic subgroups. We further link report-level quality to condition-level diagnostic accuracy. Collectively, these analyses underscore the universality, robustness, and clinical alignment of UNIRG-CXR.

Universal across institutions and data distributions Radiology reporting practices vary widely across institutions, regions [8], and documentation conventions [12]. Prior systems often overfit to dataset-specific phrasing or reporting styles, leading to performance drops when evaluated on unseen datasets or across sites [9, 15, 21]. UNIRG-CXR overcomes this limitation by exhibiting consistent high performance across all benchmark datasets and on entirely unseen distributions with zero-shot inference, including both public and private cohorts collected from diverse institutions. These results indicate that UNIRG-CXR captures the underlying clinical semantics rather than memorizing superficial textual templates—achieving true generalization across data sources, institutions, and domains. Such robustness establishes UNIRG-CXR as a universal foundation model for radiology reporting, capable of maintaining reliability and fidelity across heterogeneous real-world environments.

Universal across evaluation metrics Traditional text-generation metrics such as BLEU or ROUGE correlate only weakly with clinical judgments [13, 23], often obscuring medically significant errors. UNIRG-CXR bridges this gap by explicitly integrating clinical error signals into its reinforcement learning reward design, aligning optimization with radiological practice rather than surface-level linguistic similarity. As

a result, it achieves strong and balanced performance across both NLG and clinically grounded metrics, representing a universal improvement across evaluation dimensions.

Universal across diagnostic levels Prior work typically assessed performance at the report level [17, 26, 32], obscuring whether models captured critical findings [14]. By incorporating disease-level assessments, we provide a finer-grained view of diagnostic fidelity. These evaluations showcase the ability of UNIRG-CXR to reflect diagnostic information, including in the long tail of rare conditions where training data are sparse.

Universal across longitudinal contexts Radiologists often rely on prior studies when interpreting new exams [2, 3, 18, 33]. While previous models primarily operated in single-study settings [32], we additionally set up the longitudinal evaluation where the model is supplied with not only current CXRs but also prior reports and images. UNIRG-CXR excels in both standard and longitudinal conditions. It effectively integrates prior images and reports to produce clinically coherent updates, achieving state-of-the-art results in temporal reasoning and demonstrating superior longitudinal modeling capability.

Universal across demographics We further validate UNIRG-CXR across stratified demographic subgroups, confirming robust and equitable performance across gender, age, and race. This fairness evaluation is critical for real-world deployment, ensuring minimal bias and consistent report quality across patient populations.

Overall, UNIRG-CXR represents a substantial advance in radiology report generation, unifying high performance across datasets, metrics, diagnostic levels, longitudinal setups, and demographic subgroups. Beyond surpassing prior systems, it embodies the principles of universality, generalizability, and clinical alignment, paving the way for reliable real-world deployment. In the longer term, augmenting this system with interactive, instruction-following capabilities and expanding to multimodal patient data (e.g., lab tests, prior imaging, and clinical notes) could further enhance its clinical utility. We anticipate that UNIRG-CXR will serve as both a strong benchmark for future research and a foundation for building reliable, assistive AI systems in radiology.

Methods

Model

UNIRG-CXR is built by fine-tuning a state-of-the-art open-source vision-language foundation model, Qwen3-VL-8B-Instruct [1] for the report generation tasks. In the sections that follow, we describe our task formulation, inference and evaluation protocols, optimization strategy, and dataset curation in detail.

Tasks

Our model is trained to generate both the ‘findings’ and ‘impression’ sections of the report for a frontal view (anterior–posterior or posterior–anterior) of the chest radiograph, which typically capture the key observations made in a study. The model receives additional contextual information, including the study indication and any available comparison text. To improve computational efficiency, each radiograph is resized to a resolution of 512×512 pixels. In routine clinical practice, radiologists frequently reference prior images and prior reports when interpreting the current study. To mirror this workflow and enhance clinical fidelity, we also supply the model with the most recent prior frontal radiograph and its associated report as supplementary inputs.

Inference and Evaluation Metrics

We follow the exact setup from ReXrank to take into account context (indication + comparison). We only use the key image path provided by ReXrank which is typically frontal view image. We keep temperature as 0 for inference. We follow the setups in ReXrank [29] to evaluate report quality using the following metrics:

BLEU-2 [16]. BLEU is a standard metric for machine translation and text generation that measures n-gram precision between generated and reference texts (0–1 scale). Following Zhang et al. [29], we report BLEU-2, which captures bigram precision.

BERTScore [7, 28]. BERTScore evaluates semantic similarity by computing cosine similarity between BERT embeddings of the generated and reference reports, providing a meaning-aware alternative to surface n-gram metrics.

SembScore [20]. SembScore is a radiology-specific metric that computes cosine similarity between 14-pathology indicator vectors produced by the CheXbert labeler for generated and groundtruth reports.

RadGraph-F1 [10] measures the overlap in clinical entities and relations extracted by RadGraph from candidate and reference reports.

1/RadCliQ-v1 [24] is the reciprocal of the RadCliQ composite metric, which aggregates BLEU, BERTScore, SembScore, and RadGraph-F1 for holistic radiology report evaluation. Because RadCliQ is originally lower-is-better, we follow ReXrank [29] and report its inverse so that higher values indicate better performance to be consistent with other scores.

RaTEScore [31] is an entity-centric metric emphasizing key medical concepts such as diagnoses and anatomical structures, while being robust to medical synonyms and negation.

CheXprompt [26] In addition to the ReXrank leaderboard metrics, we also evaluate report quality with CheXprompt, an LLM-based error detection metric.

Optimization

UNIRG-CXR follows a two-stage SFT + RL training pipeline as we found that the combined approach is better than SFT alone or RL alone (details are in the supplementary information section).

In the **SFT stage**, the model is initialized with a strong foundation for radiology report generation. The SFT training is performed over four datasets: MIMIC, CheXpert-Plus, ReXGradient and IU and we conduct a grid search over learning rates $[1 \times 10^{-5}, 5 \times 10^{-5}]$ and batch sizes [128, 256, 512]. The optimal configuration is a learning rate of 5×10^{-5} with a batch size of 256 for 3 epochs.

In the **RL stage**, we adopt GRPO [19] as our reinforcement learning algorithm, which eliminates the need for value functions by computing advantages within query-specific groups. Following recent advances [25], we incorporate two key improvements: (1) a higher clipping threshold to encourage response diversity and prevent entropy collapse; (2) remove KL penalty. Our training adopts a learning rate of 5×10^{-6} , a global batch size of 256, and 16 sampled rollouts per query. The RL training is performed over four datasets: MIMIC, CheXpert-Plus, ReXGradient and IU. For reward optimization in the RL stage, we target BLEU, BERTScore, RadGraph-F1, SembScore, and CheXprompt (LLM-based), without using any format rewards since the model reliably produces well-structured reports after SFT. Our RL procedure follows a two-step optimization strategy.

Step 1: RadCliQ-oriented optimization. We first optimize a weighted composite reward consisting of BLEU, BERTScore, SembScore, and RadGraph-F1, using the RadCliQ coefficients [24] of 0, 0.370, 0.253, and 0.377, respectively. This stage runs for one epoch and encourages the model to generate outputs that are both lexically and clinically aligned with the ground-truth reports, effectively targeting RadCliQ improvement.

Step 2: Error-reduction optimization. Starting from the best checkpoint from Step 1’s RadCliQ optimization, we perform an additional epoch in which we incorporate the CheXprompt error metric into the reward. Specifically, we use $1/(\# \text{ CheXprompt errors} + 1)$ as the CheXprompt reward to incentivize reducing factual reporting errors. To preserve the RadCliQ performance achieved in Step 1, we integrate the CheXprompt reward with a weight of 0.5 alongside the previous metrics. We also apply a KL regularization term with coefficient 0.03 to prevent excessive deviation from the Step 1 policy.

Dataset Details

Our training data consists of the training splits from MIMIC-CXR, CheXpert Plus, ReXGradient and IU. We extracted the indication, comparison, findings, and impression sections from the corresponding radiology reports. We then removed studies in which both the findings and impression sections were empty. Studies that contained a findings section but lacked an impression, or contained an impression but lacked findings, were retained. Depending on which ground-truth sections were available (findings and/or impression), we

applied different prompt templates, as shown in the supplementary information section. We also set aside 1,024 samples from the MIMIC-CXR training set as a validation set for all experiments.

We follow ReXrank evaluation to evaluate UNIRG-CXR on ReXrank official test sets from MIMIC-CXR, CheXpert Plus, IU-Xray and ReXGradient private test set. Apart from the ReXrank testsets, we also evaluate on a proprietary dataset which we name as PD. Below are the details for each dataset.

MIMIC-CXR [11]. A large, publicly available dataset containing 377,110 chest X-rays from 227,835 studies collected at the Beth Israel Deaconess Medical Center in Boston, MA. All images and reports are fully de-identified. We use the official ReXrank MIMIC test set, which includes 2,347 studies, for evaluation, and use the remaining training split for model development.

CheXpert Plus [4] A publicly available dataset comprising 223,462 paired radiology reports and chest x-rays from 187,711 studies across 64,725 patients. We adopt the ReXrank test set, which follows the official CheXpert Plus test split and contains 200 studies. The CheXpert Plus training split is used for training UNIRG-CXR.

ReXGradient [30] A large-scale dataset curated by GradientHealth, consisting of a private testset of 10,000 studies from 7,004 patients across 67 clinical sites in the United States. The publicly released official training set, comprising 140,000 studies, is used for training UNIRG-CXR.

IU-Xray [6] A public dataset containing 7,470 radiology reports paired with corresponding frontal and lateral chest x-rays. We follow the ReXrank split and evaluate on the test set of 590 studies and train with the rest as the train set. To assess the zero-shot generalization capability of UNIRG-CXR on unseen data sources, we exclude the IU-Xray training set from the training corpus in the generalization study.

Proprietary Dataset (PD) This proprietary dataset comprises 11,815 chest X-ray studies from inpatient and outpatient facilities across the United States. Each study includes corresponding radiology reports of frontal and lateral view images, without prior studies. The dataset was used exclusively for evaluation, with no overlap with any training sources.

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Supplementary Information

Comparing SFT and RL

To identify the most effective training strategy for radiology report generation, we compare three approaches on the MIMIC dataset: SFT alone, RL alone, and the combined SFT + RL pipeline. All SFT and RL runs are trained for 3 epochs, while the SFT + RL configuration applies an additional 2 epochs of RL starting from the SFT checkpoint. As shown in Table 1, RL alone outperforms SFT on BERTScore and SembScore, but lags significantly behind on RadGraph-F1. This is likely because RadGraph is more sensitive to specific wording, phrasing, and lexical distributions, which SFT captures more directly through next-token prediction. Consequently, SFT slightly outperforms RL on the overall RadCliQ score. The combined *SFT + RL* setup yields the strongest results across all metrics. SFT first teaches the model the output format and dataset-specific lexical structure, and RL subsequently refines this foundation by optimizing more semantically and clinically aligned rewards, leading to the best synergy and overall performance.

Table 1: Comparison of SFT, RL, and SFT + RL on findings + impression generation on MIMIC. The combined SFT + RL strategy achieves the best performance.

Method	BERTScore	SembScore	RadGraph-F1	1/RadCliQ
Baseline	0.293	0.242	0.102	0.625
MIMIC SFT	0.421	0.426	0.236	0.962
MIMIC RL	0.437	0.435	0.198	0.950
MIMIC SFT + MIMIC RL	0.449	0.478	0.267	1.110

Prompt Templates

Below are training and inference prompt templates for findings + impression generation and findings generation. {input} is the placeholder for the input images and the context from the indication and comparison sections.

Findings + Impression Generation Prompt

This is a radiology report generation task. Here is the context: {input} Given the image and the context, provide the report in the following format: Findings: [write the findings] Impression: [write the impression] Now write the report in the format above.

Findings Generation Prompt

This is a radiology report generation task. Here is the context: {input} Given the image and the context, provide the findings in the following format: Findings: [write the findings] Now write the report in the format above.